Assessing the Threat of Adversarial Examples on Deep Neural Networks for Remote Sensing Scene Classification: Attacks and Defenses

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***Abstract*— Deep neural networks, which can learn the representative and discriminative features from data in a hierarchical manner, have achieved state-of-the-art performance in the remote sensing scene classification task. Despite the great success that deep learning algorithms have obtained, their vulnerability toward adversarial examples deserves our special attention. In this article, we systematically analyze the threat of adversarial examples on deep neural networks for remote sensing scene classification. Both targeted and untargeted attacks are performed to generate subtle adversarial perturbations, which are imperceptible to a human observer but may easily fool the deep learning models. Simply adding these perturbations to the original high-resolution remote sensing (HRRS) images, adversarial examples can be generated, and there are only slight differences between the adversarial examples and the original ones. An intriguing discovery in our study shows that most of these adversarial examples may be misclassified into the wrong category by the state-of-the-art deep neural networks with very high confidence. This phenomenon, undoubtedly, may limit the practical deployment of these deep learning models in the safety-critical remote sensing field. To address this problem, the adversarial training strategy is further investigated in this article, which significantly increases the resistibility of deep models toward adversarial examples. Extensive experiments on three benchmark HRRS image data sets demonstrate that while most of the well-known deep neural networks are sensitive to adversarial perturbations, the adversarial training strategy helps to alleviate their vulnerability toward adversarial examples.**

***Index Terms*— Adversarial attack, adversarial defense, adversarial example, deep learning, remote sensing, scene classification.**

I. Introduction

With the rapid development of remote sensing imaging technology, plenty of high-resolution remote sensing (HRRS) images are now available [1]–[4]. The detailed spatial information contained in HRRS images provides the researchers with an opportunity to observe the Earth more accurately [5]–[7]. Scene classification is the fundamental task for HRRS image interpretation, which aims to classify the extracted subregions of HRRS images into a discrete set of meaningful land use and land cover (LULC) categories [8]. Due to the high complexity of the ground objects, similar land-cover types may be shared in different scenes. For example, both residential and commercial areas may consist of roads, buildings, and vegetation objects. The main discrepancy between these two typical scene classes lies in the spatial distribution of the aforementioned objects, which is obvious for a human observer but may bring about the difficulty for automatic identification algorithms. This challenge is also known as the "semantic gap" between the low-level features and the high-level semantic concepts [9], [10].

The early research for the remote sensing scene classification task mainly focuses on handcrafted features, such as shape, texture, color, spatial and spectral features [11]–[18]. In [15], several well-known color descriptors, including global color histogram (GCH) [19], color coherent vector (CCV) [20], and autocolor correlation (ACC) [21], are evaluated for both remote sensing image retrieval and scene classification tasks in detail. The main advantage of these color descriptors is their invariance toward image translation and rotation since they focus more on the color statistics of the image from a global perspective. However, when confronted with images having similar gray values with a different color distribution, it would be very difficult for these methods to achieve a high accuracy [22]. Texture descriptors, on the other hand, characterize the image textures and spatial structure information by observing the region homogeneity and the histograms of the image subregions. Typical approaches include gray-level co-occurrence matrix (GLCM) [23], Gabor features [24], and local binary patterns (LBPs) [25]. Luo *et al.* [26] proposed to combine multiple features extracted from images with different spatial resolutions in the scene classification. The wavelet transform [27], Gabor filtering, and scale-invariant feature transform (SIFT) [28] are conducted for the feature fusion. Their experimental results show that the combinations of complementary features can further improve the classification performance [26]. This conclusion is also supported by Avramović and Risojević [29], where the gist [30] and SIFT descriptors are combined together for remote sensing scene classification.

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Fig. 1. Illustration of attacks on deep learning models for remote sensing scene classification. (Top row) Original HRRS images. (Middle row) Subtle adversarial perturbations, which are imperceptible to a human observer. (Bottom row) Generated adversarial images, which can fool the deep neural network to make wrong predictions with very high confidence.

In contrast to the aforementioned methods that directly adopt the low-level visual attributes in the classification, the midlevel scene classification approaches aim to construct a holistic scene representation with the high-order statistical patterns [31]–[39]. A general midlevel framework is to first extract handcrafted features with the aforementioned methods, and then encode or quantize these features to form a holistic representation for HRRS images [40]. One of the most representative midlevel approaches is the bag-of-visual-words (BoVW) model [41]. The visual words are commonly generated by the unsupervised clustering methods (e.g., k-means clustering) using the extracted low-level features, which are subsequently fed into the supervised classifiers for scene classification. Probabilistic topic model (PTM) is another useful midlevel approach [42]–[45]. Although the BoVW model records the histogram or frequency of visual words in the image, the PTM-based methods regard the whole image as a mixture of different topics. Zhu *et al.* [45] proposed a sparse homogeneous–heterogeneous PTM model, which explores the effect of both the homogeneous and heterogeneous information for scene classification. Several different types of low-level features and latent topics are fused with the proposed sparse topic inference procedure.

Witnessing the great success of deep learning algorithms in the remote sensing field [46]–[58], recent research for remote sensing scene classification mainly focuses on the deep feature extraction [59]–[66]. Compared with low-level and midlevel approaches which largely depend on the handcrafted features, deep learning algorithms can learn the representative and discriminative features from the image in a hierarchical manner without prior knowledge from the experts [67], [68]. Zhang *et al.* [59] proposed a novel deep unsupervised feature learning framework for scene classification. A representative set of patches from the salient regions in the image is first extracted via the saliency detection algorithm. Then, the sparse autoencoder (AE) is utilized to learn hierarchical features, which are classified using the support vector machine (SVM) classifier. Apart from AE, the convolutional neural network (CNN)-based methods deserve special attention, because of their strong feature representation ability. Castelluccio *et al.* [60] utilized the-state-of-the-art CNNs like CaffeNet [69] and GoogLeNet [70] to classify the remote sensing scenes with different training modalities. Their experimental results reveal that the CNN models pretrained on the large natural image data sets (e.g., the ImageNet data set [71]) can generalize well to remote sensing scenes.

Despite the great success that deep learning algorithms have obtained, their vulnerability toward adversarial examples cannot be neglected [72], [73]. A good review of this research topic can be found in [74]. Szegedy *et al.* [75] first discovered that deep neural networks are very susceptible to adversarial attacks (i.e., to fool the model with perturbed data) using box-constrained L-BFGS algorithm. Goodfellow *et al.* [76] showed that the primary cause of neural networks' vulnerability to adversarial perturbation is their linear nature. Based on this theory, they proposed a gradient-based approach to generate adversarial perturbation, named the fast gradient sign method (FGSM). Moosavi-Dezfooli *et al.* [77] proposed the DeepFool algorithm to efficiently compute perturbations that fool deep models, and thus reliably quantify the robustness of these classifiers. Moosavi-Dezfooli *et al.* [78] further found that the existence of universal adversarial perturbations by adding very small perturbation vectors to natural images can cause misclassification for different deep neural networks with high probability. Although these adversarial examples may remain imperceptible to a human observer, they can easily fool the deep learning models to yield wrong predictions [77], [79], [80].

So far, the threat of adversarial examples on deep neural networks for remote sensing community has not been fully explored yet. Compared with the training data used for the traditional image recognition task (e.g., the ImageNet data set [71]), the labeled samples used for remote sensing scene classification are relatively limited [22]. Under this circumstance, the trained deep neural networks are more likely to suffer from the overfitting problem, resulting in a weaker generalization capability and more sensitivity to noise. Therefore, their vulnerability to adversarial examples may be even more serious. An example of adversarial attacks on deep learning models for remote sensing scene classification is shown in Fig. 1. It can be observed from Fig. 1 that although there exists very little difference between the adversarial images and the original ones for the human vision system, they can fool the deep neural network with very high confidence. This phenomenon, undoubtedly, may limit the practical deployment of these deep learning models in the safety-critical remote sensing field. Czaja *et al.* [81] conducted a preliminary analysis about adversarial examples for remote sensing data. They emulated physical attacks by putting a $n \times n$ patch in the center of a remote sensing image to fool the deep neural network.

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Nevertheless, their work considers only the targeted attacks. Whether untargeted attacks would also bring about a threat to remote sensing image interpretation, and how to resist such attacks are not discussed in their work.

To address the aforementioned issues, in this article, we systematically analyze the threat of adversarial examples on deep neural networks for remote sensing scene classification. The main contributions of this article are summarized as follows.

1) We systematically introduced the concept of adversarial attack and defense into the remote sensing community for the first time, which reveals the significance of the resistibility of deep neural networks when addressing the safety-critical remote sensing tasks.
2) Both targeted and untargeted attacks are explored for remote sensing scene classification. Experiments on three benchmark data sets show that most of the adversarial examples generated by these attacks may be misclassified into the wrong category by the state-of-the-art deep neural networks with very high confidence.
3) To alleviate the vulnerability of deep models toward adversarial examples, the adversarial training strategy is investigated. The experimental results demonstrate that the adversarial training is a simple but effective way to increase the resistibility of deep models toward adversarial examples for the remote sensing scene classification task.
4) The transferability of adversarial examples in remote sensing scene classification task is further investigated in this article. The experimental results reveal that the adversarial examples generated by a specific deep neural network may also be detrimental to other deep models or even traditional shallow classifiers.

The rest of this article is organized as follows. Sections II and III describe the adversarial attacks and defenses in detail, respectively. Section IV presents the information on data sets used in this article and the experimental results. Conclusions and other discussions are summarized in Section V.

II. Adversarial Attack

A. Box-Constrained L-BFGS

Let $\{x_i\}_{i=1}^m$ be a set of remote sensing images and $\{y_i\}_{i=1}^m$ be the corresponding labels. We denote a classifier mapping that maps an image vector to a discrete label set by $f : x \in \mathbb{R}^n \to y \in \mathbb{L}$. Note that f is modeled by a deep CNN in this article. Then, for a given image x and a target label $\hat{y}$, in order to generate a subtle adversarial perturbation ρ that can cheat f, we can solve the following box-constrained optimization problem:

$$\text{objective: } \min_{\rho} \|\rho\|_2, \quad \text{s.t.} : \begin{cases} f(x+\rho) = \hat{y} \\ x + \rho \in [0,1]^n. \end{cases} \tag{1}$$

Note that only when $\hat{y} \neq y$, where y denotes the true label of x, (1) is nontrivial (otherwise, we will get $\rho = 0$). Since the minimizer of (1) is not unique, Szegedy *et al.* proposed to approximate the solution using a box-constrained L-BFGS [75]. More specifically, they perform the line search to find the minimum $c > 0$ for which the minimizer ρ of the following problem satisfies $f(x+\rho) = \hat{y}$, where $\hat{y} \neq y$:

$$\begin{aligned} \text{objective} : \min_{\rho} c\|\rho\|_2 + J(\theta, x+\rho, \hat{y}) \\ \text{s.t.} : x + \rho \in [0,1]^n \end{aligned} \tag{2}$$

where θ denotes the parameters in f, and $J(\theta, x+\rho, \hat{y})$ is the associated continuous loss function of f (e.g., the cross-entropy function). In fact, (2) can be interpreted as finding an adversarial image $x+\rho$, which is misclassified by f and has the closest distance to the original image x.

B. FGSM

The optimization of the problem shown in (2) is rather difficult even for a shallow neural network in practice since it involves optimizing the parameters in each layer using the adversarial examples generated from previous layers [73]. To implement efficient adversarial attacks, Goodfellow *et al.* [76] proposed a gradient-based approach to generate adversarial perturbation, named FGSM. Given a vectorized image x and its true label y, the adversarial perturbation ρ and the corresponding adversarial example x_{adv} can be calculated as

$$\begin{cases} \rho = \epsilon \cdot \text{sign}\left(\dfrac{\partial J(\theta, x, y)}{\partial x}\right) \\ x_{\text{adv}} = x + \rho \end{cases} \tag{3}$$

where $\text{sign}(\cdot)$ is the sign function, and ϵ denotes a small scalar value that restricts the norm of the perturbation. Since $|\text{sign}(x)| \leq 1$ is satisfied for any input x, the infinite norm of the generated adversarial perturbation $\|\rho\|_\infty$ thereby is no more than ϵ. This property guarantees that the difference between the adversarial example $x+\rho$ and the original image x will be imperceptible given a small enough ϵ.

In fact, FGSM can be interpreted as a gradient ascent method [76]. Given a deep neural network f with fixed parameters θ, to minimize the loss function $J(\theta, x, y)$ into the case where the predicted label of the input image x is closer to its true label y, we can regard x as a trainable variable and apply the gradient descent algorithm to x

$$x' = x - \epsilon \cdot \frac{\partial J(\theta, x, y)}{\partial x} \tag{4}$$

where x' is the updated image, and ϵ here acts as the learning rate. We can further modify (4) with the sign function as

$$x' = x - \epsilon \cdot \text{sign}\left(\frac{\partial J(\theta, x, y)}{\partial x}\right). \tag{5}$$

Note that the sign function here only changes the magnitude of the gradient but keeps the gradient direction. Thus, (4) and (5) can lead to similar effect.

We can find that the subtracted item in (5) is actually equivalent to ρ in (3). Therefore, $x+\rho$ is the opposite form of the gradient descent algorithm shown in (5), which will make the predicted label farther away from the true label y.

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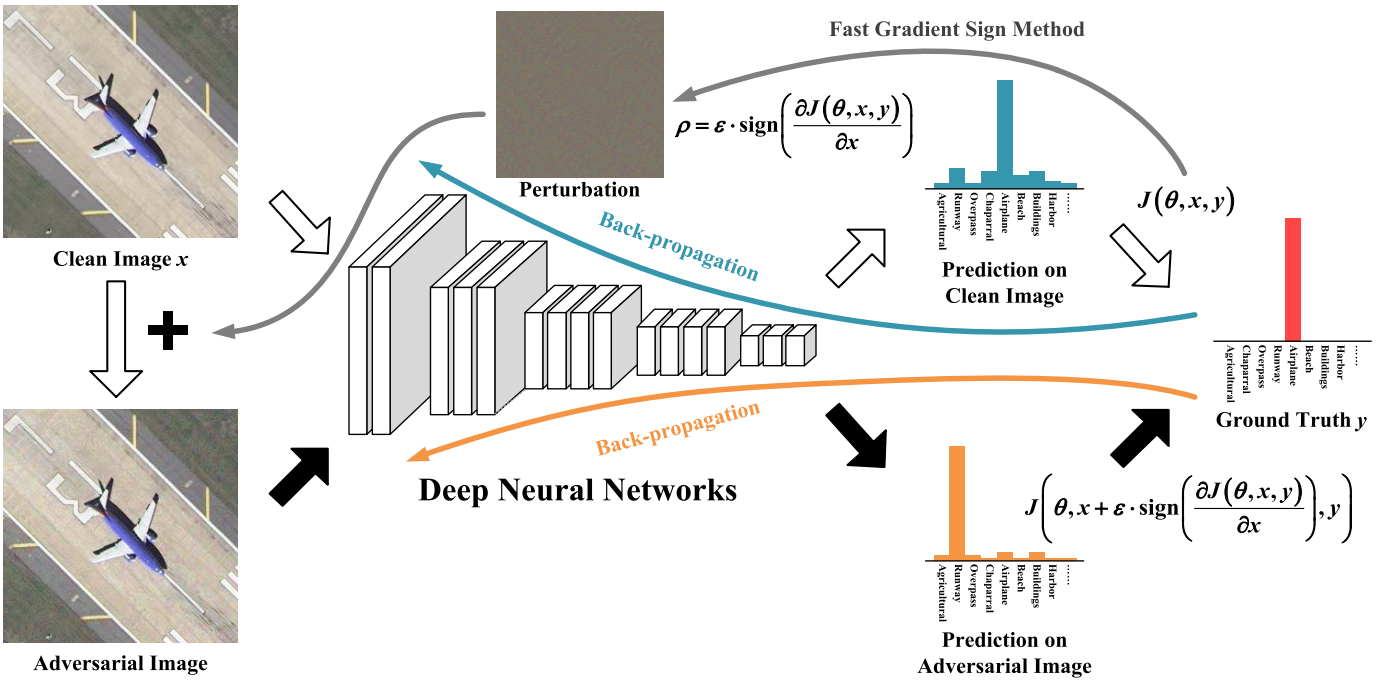


Fig. 2. Illustration of the adversarial training for remote sensing scene classification. The original clean images (denoted by hollow arrows) are first used to train a deep neural network. Then, the adversarial images (denoted by solid arrows) are generated by FGSM. The adversarial training is achieved by simultaneously minimizing the cross-entropy loss from both the original clean images and the adversarial images.

C. Targeted and Untargeted Attacks

Different from the box-constrained L-BFGS method that aims to transform the prediction of the adversarial example x_{adv} from the original true label y into the target label $\hat{y}$, FGSM shown in (3) does not restrain that x_{adv} should be misclassified into a specific category. Instead, it only attempts to cheat the deep neural network f to make a wrong prediction. Therefore, FGSM is known as an untargeted attack approach. We can also change the original FGSM into a targeted form without much effort. Given a vectorized image x and a target label $\hat{y}$, (3) can be modified as

$$\begin{cases} \rho = \epsilon \cdot \mathrm{sign}\left(\dfrac{\partial J(\theta, x, \hat{y})}{\partial x}\right) \\ x_{\mathrm{adv}} = x - \rho. \end{cases} \tag{6}$$

Note that the generated perturbation ρ, in this case, should be subtracted from the original image x to obtain the adversarial example x_{adv}.

Both (3) and (6) are one-step methods that only take a single step to fool the deep neural network f. An intuitive extension of these methods is to take multiple steps and update the adversarial example iteratively. The iterative version of FGSM can be calculated as follows:

$$\text{targetd}: x_{\mathrm{adv}}^{i+1} = x_{\mathrm{adv}}^{i} - \epsilon \cdot \mathrm{sign}\left(\frac{\partial J\left(\theta, x_{\mathrm{adv}}^{i}, \hat{y}\right)}{\partial x_{\mathrm{adv}}^{i}}\right) \tag{7}$$

$$\text{untargetd}: x_{\mathrm{adv}}^{i+1} = x_{\mathrm{adv}}^{i} + \epsilon \cdot \mathrm{sign}\left(\frac{\partial J\left(\theta, x_{\mathrm{adv}}^{i}, y\right)}{\partial x_{\mathrm{adv}}^{i}}\right) \tag{8}$$

where x_{adv}^{i} denotes the generated adversarial example at the ith iteration, and x_{adv}^{0} is initialized with the original clean image x.

III. Adversarial Defense

Stability and reliability are important factors when addressing remote sensing tasks since many of these tasks are highly related to national defense security. Although deep neural networks have shown their great capability in many challenging remote sensing tasks, their vulnerability toward adversarial examples has not received enough attention. Fig. 1 demonstrates that although there exists very little difference between the adversarial examples and the original remote sensing images for the human visual system, they can fool the deep neural network to make wrong predictions with very high confidence. This phenomenon, undoubtedly, may limit the practical deployment of these deep learning models in the safety-critical remote sensing field. An intuitive idea to address this problem is to train the deep neural network directly with the generated adversarial examples so that the network can learn to resist the attack from the adversarial examples and become more robust. In this section, we introduce the adversarial training strategy as a simple but effective adversarial defense method in detail.

As shown in Fig. 2, the main idea of the adversarial training strategy is to train the deep neural network with the mixture of both the original clean remote sensing images and the generated adversarial examples. More specifically, given a set of clean images $X = \{x_1, x_2, \ldots, x_m\}$ and the corresponding label set $Y = \{y_1, y_2, \ldots, y_m\}$, we can first optimize a deep neural network f with a continuous loss function $J(\theta, x, y)$

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using the minibatch stochastic gradient descent algorithm. In this article, we implement the function $J(\theta, x, y)$ with the cross-entropy as follows:

$$J(\theta, x, y) = y \log(f(x)) + (1 - y) \log(1 - f(x)). \quad (9)$$

Then, for each sample in X, we can generate its adversarial example x_{adv} with FGSM, and get the corresponding adversarial training set X_{adv}. The goal of adversarial training is to optimize f into the case where both the clean image x and the corresponding adversarial example x_{adv} are assigned with the same true label y. The complete loss function $\mathcal{L}_{\text{adv}}(\theta, x, y)$ can be defined as

$$\mathcal{L}_{\text{adv}}(\theta, x, y) = J(\theta, x, y) + J\left(\theta, x + \epsilon \cdot \text{sign}\left(\frac{\partial J(\theta, x, y)}{\partial x}\right), y\right). \quad (10)$$

The detailed steps of the adversarial training strategy are given in Algorithm 1.

Algorithm 1 Adversarial Training Strategy

Input:

1) A set of clean images $X = \{x_1, x_2, \ldots, x_m\}$ and the corresponding labels $Y = \{y_1, y_2, \ldots, y_m\}$.
2) A deep neural network f with parameters θ.
3) A small scalar value ϵ that restricts the norm of the generated perturbation.

Let θ_0 be the initial parameters, τ be the expected training epochs, and η be the learning rate.
$\theta \leftarrow \theta_0$
$t \leftarrow 0$
while $t < \tau$ **do**
  Compute $\mathcal{L} = \sum_{i=1}^{m} J(\theta, x_i, y_i)$ via (9).
  $\theta \leftarrow \theta - \eta \frac{\partial \mathcal{L}}{\partial \theta}$
  $t \leftarrow t + 1$
end while
$t \leftarrow 0$
while $t < \tau$ **do**
  Compute $\mathcal{L} = \sum_{i=1}^{m} \mathcal{L}_{\text{adv}}(\theta, x_i, y_i)$ via (10).
  $\theta \leftarrow \theta - \eta \frac{\partial \mathcal{L}}{\partial \theta}$
  $t \leftarrow t + 1$
end while

Output: An adversarial trained deep neural network f.

IV. Experiments

A. Data Descriptions

Three benchmark HRRS image data sets, including the UC Merced (UCM) data set [41], the WHU-RS19 data set [82], and the AID data set [40], are utilized in this article. Some example images from these three data sets are presented in Figs. 3–5.

1) UCM Data Set: This data set consists of 2100 overhead scene images including 21 land-use classes. Each class contains 100 aerial images measuring 256 × 256 pixels, with a spatial resolution of 0.3 m/pixel in the red–green–blue color space. This data set is extracted from aerial orthoimagery downloaded from the U.S. Geological Survey (USGS) National Map of the following U.S. regions: Birmingham, Boston, Buffalo, Columbus, Dallas, Harrisburg, Houston, Jacksonville, Las Vegas, Los Angeles, Miami, Napa, New York, Reno, San Diego, Santa Barbara, Seattle, Tampa, Tucson, and

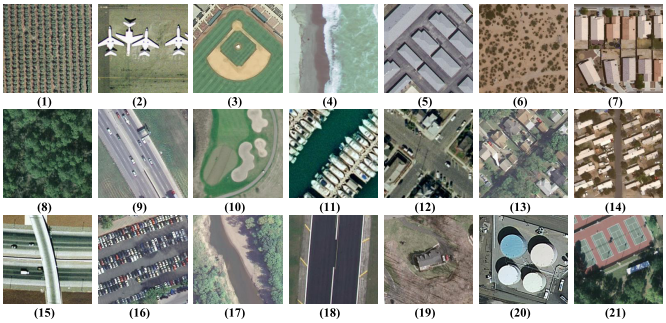


Fig. 3. Example images from the UCM data set. 1: agricultural. 2: airplane. 3: baseball diamond. 4: beach. 5: buildings. 6: chaparral. 7: dense residential. 8: forest. 9: freeway. 10: golf course. 11: harbor. 12: intersection. 13: medium residential. 14: mobile home park. 15: overpass. 16: parking lot. 17: river. 18: runway. 19: sparse residential. 20: storage tanks. 21: tennis court.

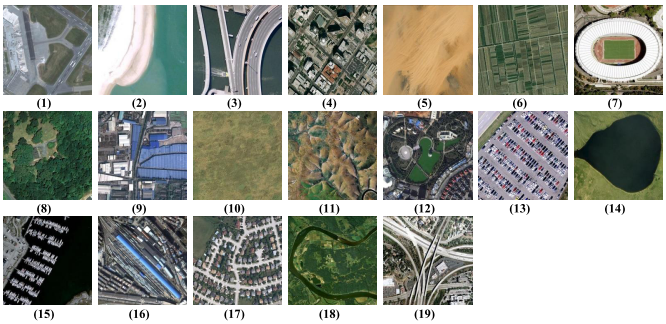


Fig. 4. Example images from the WHU-RS19 data set. 1: airport. 2: beach. 3: bridge. 4: commercial. 5: desert. 6: farmland. 7: football field. 8: forest. 9: industrial. 10: meadow. 11: mountain. 12: park. 13: parking. 14: pond. 15: port. 16: railway station. 17: residential. 18: river. 19: viaduct.



Fig. 5. Example images from the AID data set. 1: airport. 2: bare land. 3: baseball field. 4: beach. 5: bridge. 6: center. 7: church. 8: commercial. 9: dense residential. 10: desert. 11: farmland. 12: forest. 13: industrial. 14: meadow. 15: medium residential. 16: mountain. 17: park. 18: parking. 19: playground. 20: pond. 21: port. 22: railway station. 23: resort. 24: river. 25: school. 26: sparse residential. 27: square. 28: stadium. 29: storage tanks. 30: viaduct.

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TABLE I

CLASSIFICATION RESULTS OF THE UCM DATA SET WITH UNTARGETED ATTACKS AND DEFENSES (BEST RESULTS ARE SHOWN IN BOLD)

Methods	Clean Test Set	Adversarial Test Set	OA Gap
AlexNet	85.75±1.16	14.58±1.95	−71.17
VGGNet-16	87.12±2.01	18.75±1.79	−68.37
ResNet-18	91.44±0.67	28.57±1.54	−62.87
ResNet-50	94.16±0.73	43.98±1.23	−50.18
ResNet-101	93.21±1.19	45.49±1.41	−47.72
DenseNet-121	93.13±0.96	43.72±1.05	−49.41
DenseNet-201	**94.67±0.45**	51.74±1.34	−42.93
AlexNet-Defense	85.95±1.43	57.92±1.78	−28.03
VGGNet-16-Defense	89.11±1.06	**71.57±1.95**	**−17.54**
ResNet-18-Defense	92.23±1.19	58.02±1.66	−34.21
ResNet-50-Defense	93.22±0.55	66.35±1.58	−26.87
ResNet-101-Defense	91.87±0.96	71.48±1.81	−20.39
DenseNet-121-Defense	94.32±1.34	67.99±1.25	−26.33
DenseNet-201-Defense	94.54±1.03	69.73±0.96	−24.82

TABLE II

CLASSIFICATION RESULTS OF THE WHU-RS19 DATA SET WITH UNTARGETED ATTACKS AND DEFENSES (BEST RESULTS ARE SHOWN IN BOLD)

Methods	Clean Test Set	Adversarial Test Set	OA Gap
AlexNet	86.00±1.81	13.04±2.12	−72.96
VGGNet-16	87.75±1.88	11.95±1.87	−75.80
ResNet-18	92.05±0.85	22.17±1.26	−69.88
ResNet-50	94.28±1.00	43.39±2.45	−50.89
ResNet-101	94.27±0.85	46.06±1.86	−48.21
DenseNet-121	91.89±1.16	36.84±2.31	−55.04
DenseNet-201	94.71±0.77	48.73±1.91	−45.97
AlexNet-Defense	86.19±1.63	53.93±3.69	−32.26
VGGNet-16-Defense	89.01±2.09	66.34±2.03	**−22.67**
ResNet-18-Defense	93.34±0.95	51.79±1.15	−41.56
ResNet-50-Defense	94.16±1.26	65.55±2.39	−28.62
ResNet-101-Defense	91.96±2.73	66.93±3.71	−25.03
DenseNet-121-Defense	95.72±0.45	70.86±1.45	−24.86
DenseNet-201-Defense	**96.00±0.72**	**71.61±2.28**	−24.39

TABLE III

CLASSIFICATION RESULTS OF THE AID DATA SET WITH UNTARGETED ATTACKS AND DEFENSES (BEST RESULTS ARE SHOWN IN BOLD)

Methods	Clean Test Set	Adversarial Test Set	OA Gap
AlexNet	84.69±1.06	10.77±1.19	−73.91
VGGNet-16	86.14±1.20	15.69±1.02	−70.45
ResNet-18	91.90±0.28	16.62±1.24	−75.28
ResNet-50	92.07±0.42	30.68±2.25	−61.39
ResNet-101	91.44±0.83	32.76±2.48	−58.67
DenseNet-121	93.50±0.33	40.56±1.41	−52.93
DenseNet-201	**93.81±0.30**	51.74±1.66	−42.08
AlexNet-Defense	81.55±0.92	60.13±1.09	−21.42
VGGNet-16-Defense	84.57±1.54	72.06±1.76	**−12.50**
ResNet-18-Defense	91.78±0.44	72.98±0.67	−18.80
ResNet-50-Defense	90.87±1.12	73.32±1.04	−17.55
ResNet-101-Defense	90.79±1.10	**75.96±0.92**	−14.83
DenseNet-121-Defense	93.34±0.37	71.65±0.83	−21.70
DenseNet-201-Defense	92.62±0.60	69.88±0.87	−23.04

Ventura. The 21 land-use classes are: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium-density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts.

2) WHU-RS19 Data Set: This data set is developed in [83] on its original version of [82], extracted from a set of satellite images exported from Google Earth (Google Inc.) with spatial resolution up to 0.5 m in the red–green–blue color space. It contains 19 scene classes, including airport, beach, bridge, commercial area, desert, farmland, football field, forest, industrial area, meadow, mountain, park, parking lot, pond, port, railway station, residential area, river, and viaduct. For each scene class, there are about 50 images, with 1005 images in total for the entire data set. The image sizes are 600×600 pixels.

3) AID Data Set: This data set is also collected from Google Earth (Google Inc.). It is made up of the following 30 aerial scene types: airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct. All the images are labeled by the specialists in the field of remote sensing image interpretation. The numbers of sample images vary a lot with different aerial scene types, from 220 up to 420. In all, the AID data set has a number of 10 000 images within 30 classes. The AID data set has multiple resolutions: the pixel resolution changes from about 8 to about 0.5 m. The size of each aerial image is fixed to be 600×600 pixels to cover a scene with various resolutions.

B. Experimental Design and Implementation Details

The experiments are conducted in three parts. The first part reports the quantitative classification results of different deep models on the adversarial examples generated by both untargeted and targeted attacks. The adversarial training strategy is further evaluated as an adversarial defense method for different deep models. The second part visualizes the generated adversarial examples and analyses how the parameter ϵ and the number of steps in FGSM influence the generated adversarial examples. Finally, the third part investigates the transferability of adversarial examples to other deep models and traditional shallow classifiers.

All the experiments in this article are randomly repeated ten times with random training and test data. In each repetition, we first randomly generate the training set from the whole data set with 20% of the labeled images. Then, the remaining samples make up the test set. The overall accuracy (OA, i.e., the number of correctly classified images divided by the number of whole test set) is utilized to quantitatively estimate different methods. Both the average value and the standard deviation are reported.

We implement the experiments with the PyTorch platform in this article. The ImageNet [71] pretrained models are adopted to initialize the weights in the networks. We use the Adam optimizer to train the networks with a learning rate of $1e-4$ and a batch size of 64. The training epochs are set as 10. The experiments are carried out with two Intel Xeon Silver 4114 2.20-GHz CPUs with 128 GB of RAM, and four NVIDIA GeForce RTX 2080 Ti GPUs.

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TABLE IV

CLASSIFICATION RESULTS OF THE UCM DATA SET WITH TARGETED ATTACKS AND DEFENSES (BEST RESULTS ARE SHOWN IN BOLD). THE TARGETED CATEGORY IN THE ATTACK IS SET AS FOREST

Methods	Clean Test Set	Adversarial Test Set	OA Gap
AlexNet	85.75±1.16	32.33±2.78	−53.42
VGGNet-16	87.12±2.01	31.14±2.41	−55.98
ResNet-18	91.44±0.67	44.79±3.07	−46.65
ResNet-50	94.16±0.73	54.33±1.60	−39.82
ResNet-101	93.21±1.19	57.15±1.80	−36.06
DenseNet-121	93.13±0.96	65.85±1.79	−27.29
DenseNet-201	94.67±0.45	66.99±1.69	−27.68
AlexNet-Defense	87.20±1.19	70.74±1.85	−16.46
VGGNet-16-Defense	88.75±1.87	79.29±3.53	−9.46
ResNet-18-Defense	92.41±1.08	77.13±1.56	−15.28
ResNet-50-Defense	93.10±1.48	76.16±1.72	−16.95
ResNet-101-Defense	91.07±2.04	78.48±1.69	−12.58
DenseNet-121-Defense	94.12±0.79	**86.77±1.31**	**−7.35**
DenseNet-201-Defense	**94.96±0.59**	84.02±1.07	−10.94

TABLE V

CLASSIFICATION RESULTS OF THE WHU-RS19 DATA SET WITH TARGETED ATTACKS AND DEFENSES (BEST RESULTS ARE SHOWN IN BOLD). THE TARGETED CATEGORY IN THE ATTACK IS SET AS BEACH

Methods	Clean Test Set	Adversarial Test Set	OA Gap
AlexNet	86.00±1.81	28.46±3.31	−57.53
VGGNet-16	87.75±1.88	21.96±3.97	−65.78
ResNet-18	92.05±0.85	35.41±2.83	−56.64
ResNet-50	94.28±1.00	56.78±3.00	−37.50
ResNet-101	94.27±0.85	58.72±2.61	−35.55
DenseNet-121	91.89±1.16	69.63±3.89	−22.26
DenseNet-201	94.71±0.77	72.49±3.46	−22.22
AlexNet-Defense	88.56±1.01	68.75±2.14	−19.81
VGGNet-16-Defense	88.15±3.16	74.17±3.67	−13.98
ResNet-18-Defense	92.99±0.71	77.62±2.25	−15.36
ResNet-50-Defense	93.94±1.04	78.46±2.21	−15.48
ResNet-101-Defense	93.14±1.40	78.41±2.28	−14.72
DenseNet-121-Defense	95.32±0.51	**91.63±0.71**	**−3.69**
DenseNet-201-Defense	**95.64±1.41**	89.01±1.66	−6.64

TABLE VI

CLASSIFICATION RESULTS OF THE AID DATA SET WITH TARGETED ATTACKS AND DEFENSES (BEST RESULTS ARE SHOWN IN BOLD). THE TARGETED CATEGORY IN THE ATTACK IS SET AS FARMLAND

Methods	Clean Test Set	Adversarial Test Set	OA Gap
AlexNet	84.69±1.06	18.03±2.00	−66.66
VGGNet-16	86.14±1.20	19.02±2.10	−67.12
ResNet-18	91.90±0.28	26.12±1.87	−65.78
ResNet-50	92.07±0.42	39.89±2.37	−52.18
ResNet-101	91.44±0.83	43.26±2.46	−48.18
DenseNet-121	93.50±0.33	51.18±1.58	−42.32
DenseNet-201	**93.81±0.30**	61.45±1.98	−32.37
AlexNet-Defense	83.09±1.04	69.44±1.72	−13.65
VGGNet-16-Defense	84.88±2.58	76.72±2.67	**−8.16**
ResNet-18-Defense	92.03±0.51	**80.87±0.81**	−11.16
ResNet-50-Defense	90.92±0.87	78.33±1.49	−12.58
ResNet-101-Defense	90.72±0.86	80.63±1.24	−10.09
DenseNet-121-Defense	93.35±0.42	79.34±0.83	−14.02
DenseNet-201-Defense	93.03±0.73	78.78±0.91	−14.24

C. Adversarial Attacks and Defenses

In this section, we report the quantitative classification results of different deep models on both the clean test set (i.e., the original test set) and the adversarial test set, with and without the adversarial training strategy. The deep models utilized in this article include AlexNet [84], VGGNet-16 [85], ResNet-18 [86], ResNet-50 [86], ResNet-101 [86], DenseNet-121 [87], and DenseNet-201 [87]. We use "-Defense" to denote that the network has been trained with the adversarial training strategy.

For both the untargeted and targeted attacks, we set the parameter ϵ and the number of steps in FGSM as 0.05 and 1 in the experiments, respectively.

1) Untargeted Attacks and Defenses: The quantitative results of the untargeted attacks and defenses are presented in Tables I–III.

It can be observed that deep neural networks can yield very good performance on the original clean test set for all three data sets. Most of the deep models utilized in this article can obtain an OA of more than 90% except for AlexNet and VGGNet-16. Besides, networks with deeper architectures generally show a trend to achieve higher accuracy. Taking the UCM data set, for example, while AlexNet (which consists of eight weights layers) gets an OA of around 85%, VGGNet-16 (which consists of 16 weights layers) obtains an OA of around 87%, and ResNet-18 (which consists of 18 weights layers) further improves the OA to around 91%. DenseNet-201 yields the most robust results, whose OA can reach more than 93% for all three data sets.

However, when confronted with the adversarial test sets generated by FGSM, all the aforementioned deep models fail to yield satisfactory performance. For example, while the OA of AlexNet is about 85% on the original clean test set for the UCM data set, it drops to only 14% on the adversarial test set. The OA gap between both sets even reaches up to 71%. A similar phenomenon can be observed from the results of other models and data sets, which undoubtedly, demonstrates the vulnerability of these deep neural networks toward adversarial examples. It is also observed that the OA gap between the clean and adversarial test sets is highly related to the depth of the models. In general, the shallower the network is the larger OA gap the model may have. It can be seen from Table I, compared to the shallow networks like AlexNet, DenseNet-201 shows better robustness toward adversarial attacks, and the OA gap of DenseNet-201 is around 42%. Even though, the OA of DenseNet-201 on the adversarial test set is only around 51%, which is still not acceptable in the practical remote sensing scene classification task.

By contrast, with the help of the adversarial training strategy, the resistibility of deep models toward adversarial examples is increased. For example, while the traditional VGGNet-16 can only obtain an OA of less than 20% on the adversarial test set for all three data sets, the accuracy gets improved to over 66% with the adversarial training. As shown in Tables I–III, most of the deep models with defense operation can achieve an OA of more than 60% on the adversarial test set, and the OA gaps are reduced to less than 30% in most cases, which demonstrates the effectiveness of the adversarial training strategy.

2) Targeted Attacks and Defenses: The quantitative results of the targeted attacks and defenses are presented in

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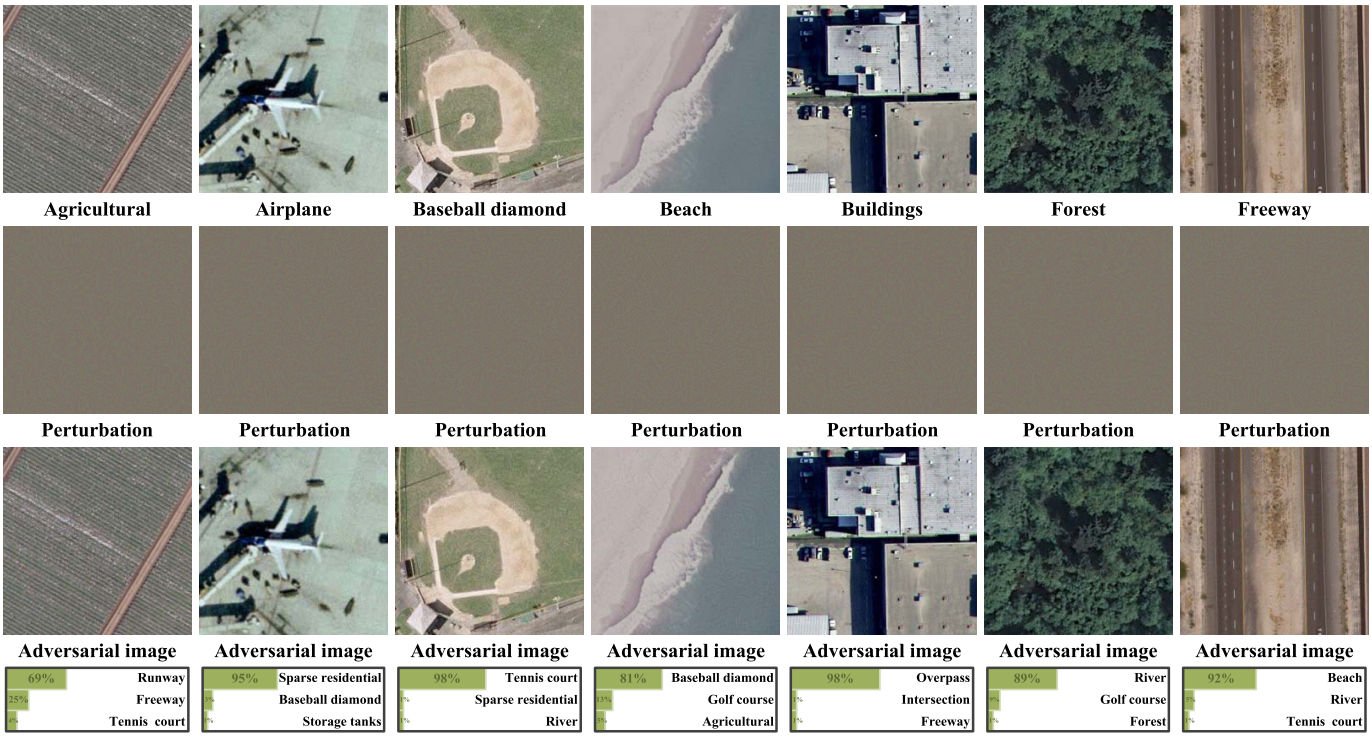


Fig. 6. Selected examples of the untargeted attacks. First row: original remote sensing images from the UCM data set. Second row: corresponding perturbation. The parameter ϵ and the number of steps in FGSM is set as 0.05 and 1 in this experiment, respectively. Third row: corresponding adversarial images. Green bars in the bottom row show the top three predicted categories for each adversarial image by AlexNet.

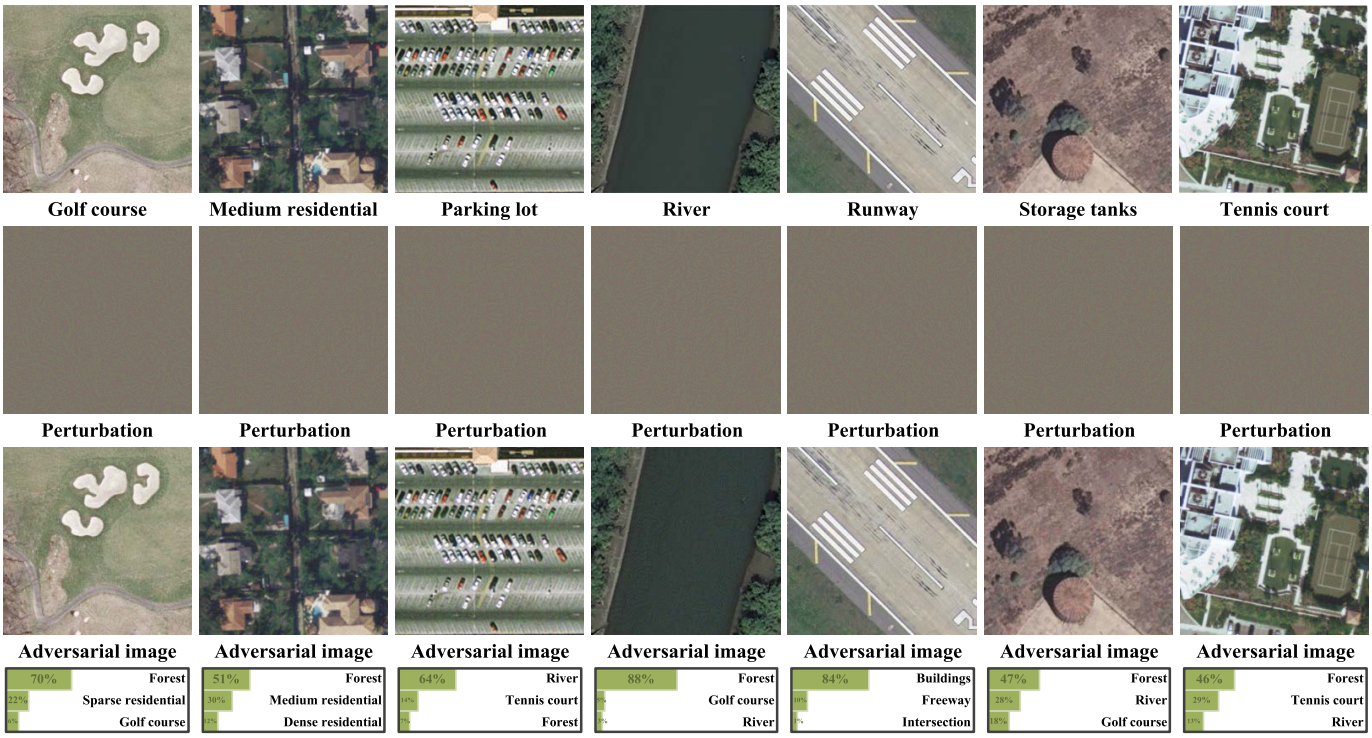


Fig. 7. Selected examples of the targeted attacks. We set the targeted category in the adversarial attack as "Forest." First row: original remote sensing images from the UCM data set. Second row: corresponding perturbation. The parameter ϵ and the number of steps in FGSM is set as 0.05 and 1 in this experiment, respectively. The third row shows the corresponding adversarial images. Green bars in the bottom row show the top three predicted categories for each adversarial image by AlexNet.

Tables IV–VI. We set the targeted categories in the attacks as "Forest," "Beach," and "Farmland," for the UCM data set, WHU-RS19 data set, and AID data set, respectively.

Similar to the untargeted attacks, targeted attacks can also fool the deep models without much effort. As shown in Table IV, while AlexNet can yield an OA about 85% on the

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original clean test set, its accuracy decreases dramatically on the targeted test set to about 32%. The corresponding OA gap is about 53%. It is observed that the influence of these targeted adversarial examples is also related to the depth of the model. Deep neural networks like ResNet and DenseNet generally get less perturbed compared with AlexNet and VGGNet.

Another interesting finding is that the untargeted attacks generally lead to larger OA gaps compared to targeted attacks. Taking the UCM data set, for example, while the OA gap of DenseNet-201 on the untargeted adversarial test set is about 42% as shown in Table I, the one on the targeted adversarial test set is about 27% as shown in Table IV. A similar phenomenon can be observed from the results of other models and data sets. One possible reason for this phenomenon may lie in the fact that there exist different levels of similarity between images from different categories. While the targeted attacks aim to force the network to yield a specific predicted label for all different images, the untargeted attacks only attempt to make the network generate a wrong prediction, without any restriction for the predicted label. Therefore, the untargeted attacks tend to fool the deep models more easily under the same level of perturbations, compared with the targeted attacks. Even though, the threat of targeted adversarial examples should not be ignored.

From the results shown in Tables IV–VI, it can be observed that the adversarial training strategy is an effective way to deal with the targeted attacks. In most cases, the OA gap between the original clean test set and the adversarial test set can be reduced to less than 15% with this defense strategy. In the experiment of WHU-RS19 data set, DenseNet-121-Defense even yields an OA of about 91% on the adversarial test set, which demonstrates a great immunity to the targeted attacks. The corresponding OA gap is only about 3%, almost showing no difference in the original clean test set and the adversarial test set.

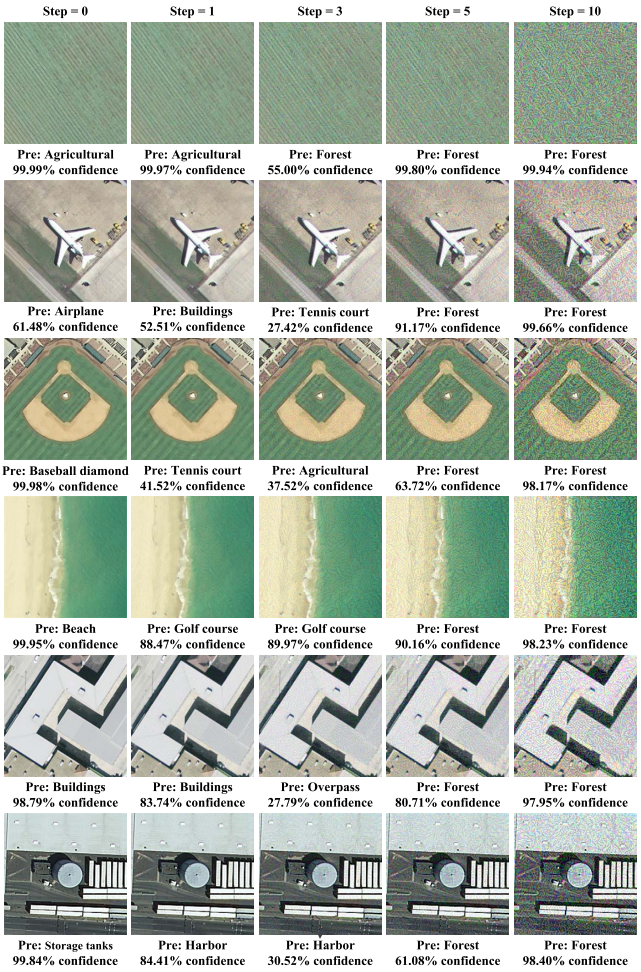


Fig. 8. Selected adversarial examples generated by the targeted attacks at different steps. We set the targeted category in the adversarial attack as "Forest." First column: original remote sensing images from the UCM data set (step = 0). The parameter ϵ in FGSM is set as 0.05 in this experiment. The top one predicted category by AlexNet along with the corresponding confidence is given beneath each image.

D. Visualization of the Adversarial Examples

In Section IV-C, we have presented the quantitative results of the adversarial attacks on deep neural networks. In this section, we further visualize the generated adversarial examples and analyze how the parameter ϵ and the number of steps in FGSM influence the generated adversarial examples.

The examples of the untargeted adversarial attacks are presented in Fig. 6. The parameter ϵ and the number of steps in FGSM is set to 0.05 and 1 in this experiment, respectively. It can be observed from the second row in Fig. 6 that the generated adversarial perturbations are very subtle and have random patterns. For a human observer, the difference between the original remote sensing images (shown in the first row) and the corresponding adversarial images (shown in the third row) is usually imperceptible. However, state-of-the-art deep neural networks can be easily fooled by these deceptive images. The top three predicted categories for each adversarial image by AlexNet are presented in the bottom row. It can be observed that the deep neural network makes wrong predictions with relatively high confidence in most cases (i.e., with a probability greater than 80% on the top one predicted category).

While the untargeted attacks only attempt to make the network generate a wrong prediction, without any restriction for the predicted label, the targeted attacks aim to force the network to yield a specific predicted label for all different images. Thus, the realization of targeted attacks is more difficult. Some selected examples of the targeted adversarial attacks are presented in Fig. 7. The parameter ϵ and the number of steps in FGSM is set to 0.05 and 1 in this experiment, respectively. We set the targeted category in the adversarial attack as "forest." It can be observed from Fig. 7 that the top one prediction's confidence is relatively low in this case. Taking the "tennis court" shown in the last column, for example, although the corresponding adversarial image is misclassified as the targeted category "forest" by the network, the confidence of this prediction is only 46%. Another interesting phenomenon is that the targeted attacks may also lead to wrong predictions different from the targeted category specified before. For the "parking lot" in Fig. 7, the top one prediction is "river" rather than the targeted category "forest."

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TABLE VII

CLASSIFICATION RESULTS OF THE UCM DATA SET WITH UNTARGETED ATTACKS USING SHALLOW CLASSIFIERS. THE RESULTS OF DENSENET-201 AND DENSENET-201-DEFENSE ARE SELECTED FOR THE COMPARISON (BEST RESULTS ARE SHOWN IN BOLD)

Methods	Clean Test Set	Adversarial Test Set	OA Gap	Running Time (s)
DenseNet-201	**94.67±0.45**	51.74±1.34	−42.93	59.59±4.93
DenseNet-201-Defense	94.54±1.03	69.73±0.96	−24.82	119.85±8.03
RandomForest (shallow features)	40.68±1.26	35.53±1.31	−5.15	45.07±0.01
GBM (shallow features)	45.43±1.92	40.39±1.69	−5.04	57.39±0.58
KNN (shallow features)	46.27±1.37	39.99±1.44	−6.27	45.02±0.01
GBDT (shallow features)	48.76±2.14	44.51±1.98	−4.25	48.50±0.52
SVM-RBF (shallow features)	49.36±2.05	46.40±1.51	**−2.96**	45.14±0.01
RandomForest (deep features)	59.52±1.45	49.37±1.45	−10.15	24.09±0.01
GBM (deep features)	62.39±1.54	49.70±1.81	−12.69	41.89±0.94
GBDT (deep features)	75.49±1.45	60.93±1.27	−14.56	28.33±0.57
KNN (deep features)	77.01±0.92	67.25±0.96	−9.76	**24.03±0.01**
SVM-RBF (deep features)	88.17±0.61	**74.52±1.28**	−13.65	24.28±0.01

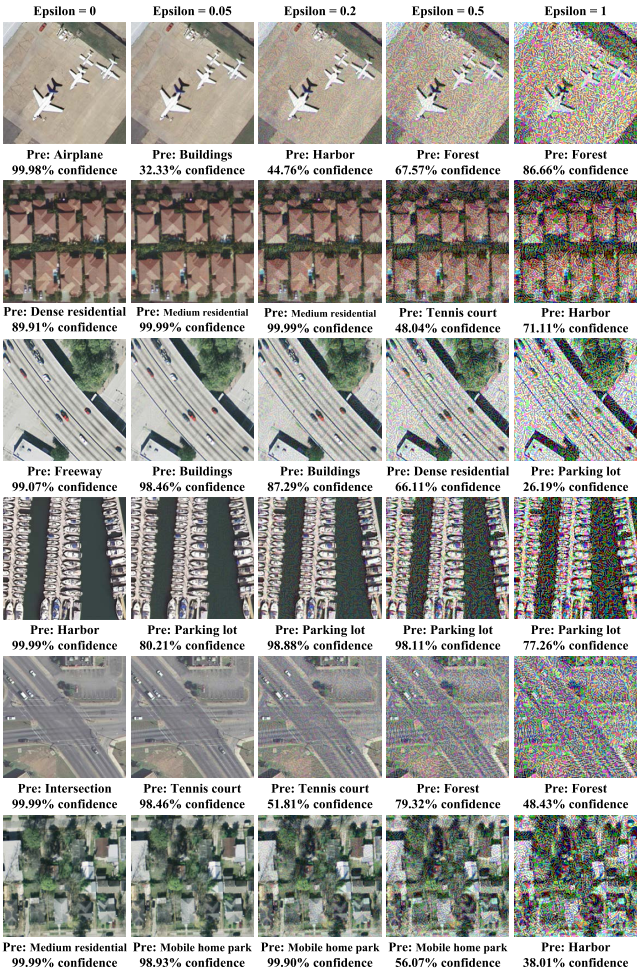


Fig. 9. Selected adversarial examples generated by the untargeted attacks with different values of ϵ. First column: original remote sensing images from the UCM data set ($\epsilon = 0$). The number of steps in FGSM is set to 1 in this experiment. The top one predicted category by AlexNet along with the corresponding confidence is given beneath each image.

One of the possible solutions to increase the success rate of the targeted attacks is to conduct multiple steps of FGSM. In Fig. 8, we present the adversarial examples generated by the targeted attacks at different steps. The parameter ϵ is fixed

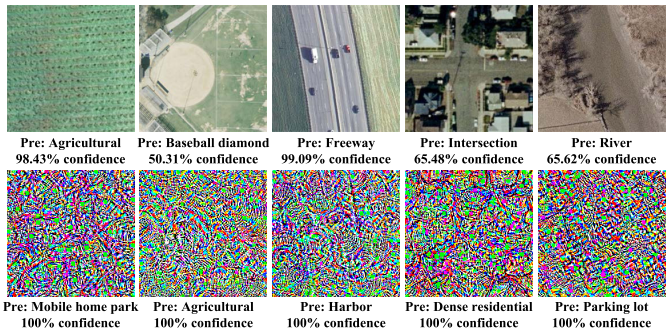


Fig. 10. Extreme cases of adversarial examples that are unrecognizable to humans, but the deep neural networks believe with 100% certainty to be a specific scene. First row: original remote sensing images from the UCM data set. Second row: Adversarial examples generated by the untargeted attacks. The parameter ϵ and the number of steps in FGSM are set as 1 and 10 in this experiment, respectively.

to 0.05 in this experiment. It can be observed that the one-step attacks are not strong enough to convert the true category into the targeted category "forest" for examples shown in Fig. 8. After we conduct five steps of FGSM, all the adversarial examples are misclassified into the "forest" category. If we further increase the number of steps to ten, all the adversarial examples can be misclassified into the targeted category with very high confidence (greater than 97%). Nevertheless, more steps of attacks may lead to a perceptible difference from the original remote sensing images, which in turn, may decrease the success rate of the attacks, since people can easily distinguish the generated adversarial examples from the real ones in this case.

The parameter ϵ is another important factor in the adversarial attacks. In Fig. 9, we analyze the influence of different values of ϵ on the generated adversarial examples. The number of steps in FGSM is fixed as 1 in this experiment. It can be observed from Fig. 9 that a large ϵ may lead to serious noises and irregular patterns in the generated adversarial examples. Besides, one-step attacks with a large ϵ may lead to unstable predictions. Taking the "freeway" shown in the third row, for example, as the ϵ becomes larger, the confidence of the top one

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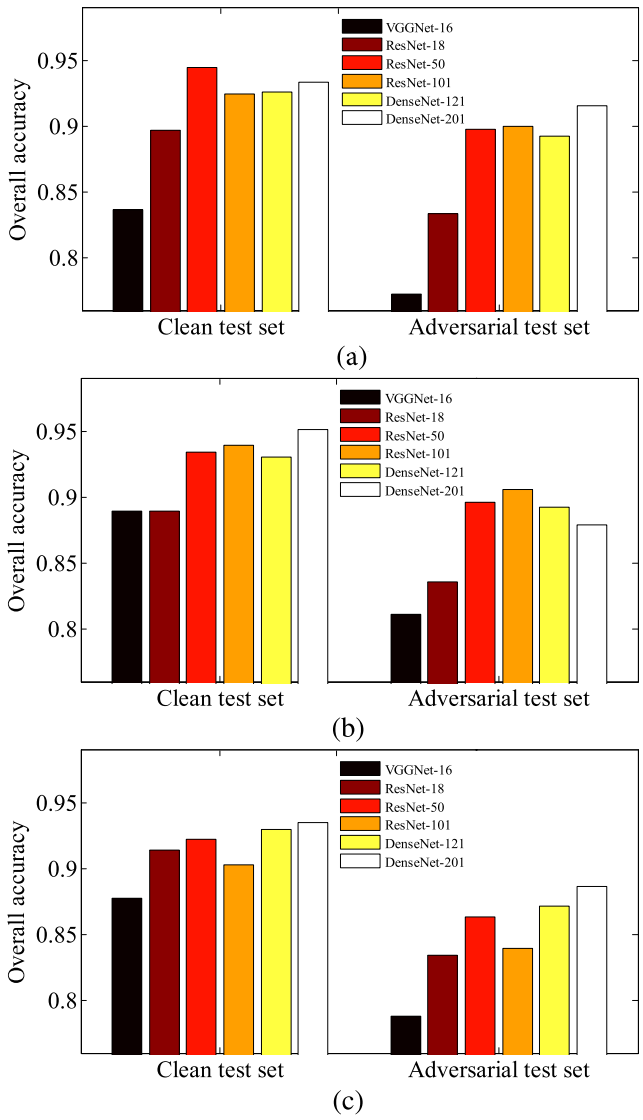


Fig. 11. Transferability of the untargeted adversarial examples generated by AlexNet. (a) UCM data set. (b) WHU-RS19 data set. (c) AID data set. The parameter ϵ and the number of steps in FGSM is set to 0.05 and 1 in this experiment, respectively.

prediction is getting lower. In the case of $\epsilon = 1$, the confidence of the top one prediction is only 26.19%.

An intriguing discovery in our study shows that if we further conduct multiple steps of FGSM with a large value of ϵ, the confidence of the top one predictions on the generated adversarial examples can reach up to 100%. Some extreme cases of adversarial examples that are unrecognizable to humans, but the deep neural networks believed to be with 100% certainty to be a specific scene are presented in Fig. 10. It can be observed from Fig. 10 that conducting ten steps of FGSM with a large value of ϵ (i.e., $\epsilon = 1$) can completely destroy the original remote sensing image. There is no similar texture, color, or structure information shared by the adversarial images and the original clean images anymore. For a human observer, the generated adversarial examples are completely random patterns with irregular noises in this case. Nevertheless, these samples successfully fool the deep neural networks into believing they belong to specific remote sensing scene categories with 100% confidence. This phenomenon also reveals the significance of the resistibility of deep neural networks when addressing the safety-critical remote sensing tasks.

E. Transferability of the Adversarial Examples

In practical applications, the detailed architecture of the deployed model may be inaccessible. Under this circumstance, we cannot directly generate adversarial examples with the deployed model. An alternative way is to use adversarial examples generated by existing deep networks to attack the target model. Thus, whether these adversarial examples are also detrimental to other deep models is worth studying. In this section, we further investigate the transferability of the adversarial examples generated by AlexNet on other deep networks. The results are presented in Fig. 11.

It can be observed that while these deep models can yield good performance on the original clean test set, their OAs on the adversarial test set generated by AlexNet drop in different degrees. This phenomenon shows that the adversarial examples generated by a specific network may also be a threat to other deep models. Besides, deeper architectures generally possess a stronger resistibility toward adversarial examples compared to shallow models. For example, in the UCM data set, the OA gap of VGGNet-16 is much larger than the one of DenseNet-201, as shown in Fig. 11(a).

Another interesting research direction is to investigate the influence of adversarial examples on traditional classification approaches. Here, we select several classical shallow classifiers including random forest, gradient boosting machine (GBM), k-nearest neighbors (KNNs), gradient boosting decision tree (GBDT), and SVM with radial basis function kernel (SVM-RBF). For each classifier, we test the classification performance on both the clean test set and adversarial test set using shallow features and deep features, respectively. The shallow features are extracted by LBPs [25] using a circle operator with a radius of 3, and the number of points in the circularly symmetric neighbor is set to 8. Then, the LBP features of each image are quantized into 256 bins to get the final histogram features. For deep features, we adopt the ImageNet [71] pretrained ResNet-18 [86] as the deep features extractor. Features from the last average pooling layer in ResNet-18 of each image with a length of 512 are used in the classification. The experimental results are shown in Table VII. Here, the adversarial test set is made up of the adversarial examples generated by DenseNet-201 with untargeted attacks. It can be observed from Table VII that shallow classifiers generally yield lower OAs on the clean test set especially with shallow features. For example, while DenseNet-201 achieves an OA of 94.67% on the clean test set, the OA of SVM-RBF with shallow features is only 49.36%. On the other hand, shallow classifiers possess much better resistibility toward adversarial examples compared to deep models. For example, the OA gap of SVM-RBF with shallow features is only 2.96%, while DenseNet-201 has an OA gap of 42.93%. With the help

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of adversarial training, DenseNet-201-Defense can decrease the OA gap to 24.82%, which, however, is still much larger than those of shallow classifiers. These results indicate that shallow classifiers like SVM are less vulnerable to adversarial examples generated by deep neural networks.

Besides, compared to the performance of shallow features, deep features generally help shallow classifiers to achieve higher OAs on both clean and adversarial test sets but also lead to relatively larger OA gaps. For instance, SVM-RBF can dramatically increase the OA on the clean test set from 49.36% to 88.17% with deep features, which demonstrates the superiority of deep feature extraction from the perspective of accuracy. Nevertheless, the OA gap increases from 2.98% to 13.65% at the same time, which indicates that deep features are more vulnerable than shallow features. We can also find that with the combination of both deep features and shallow classifiers, SVM-RBF achieves the highest OA of 74.52% on the adversarial test set, which even outperforms DenseNet-201-Defense by around 5%. These results also indicate that combining the advantages of both shallow classifiers and deep features may lead to more robust performance when dealing with adversarial attacks.

We have also reported the running time of different methods in Table VII. It can be observed from Table VII that conducting adversarial defense will significantly increase the computation burden. Although the running time of DenseNet-201 is around 59 s, DenseNet-201-Defense costs about 119 s, which is approximately twice as much as the one of DenseNet-201. Compared to deep models like DenseNet-201, shallow classifiers like SVM generally costs less running time, especially with deep features, since we directly use the ImageNet pretrained model to extract deep features without any training process.

V. Conclusion

Stability and reliability are important factors when addressing remote sensing tasks since many of these tasks are highly related to national defense security. In this article, we systematically analyze the threat of adversarial examples on deep neural networks for remote sensing scene classification. Although deep learning models have achieved great success in remote sensing scene classification task, experimental results in this article show that the state-of-the-art deep neural networks can be easily fooled by adversarial examples. To alleviate the vulnerability of deep models toward adversarial examples, the adversarial training strategy is further investigated in this article. The experimental results demonstrate that the adversarial training is a simple but effective way to increase the resistibility of deep models toward adversarial examples for the remote sensing scene classification task. We also investigate the transferability of adversarial examples in remote sensing images. The experimental results reveal that the adversarial examples generated by a specific deep neural network may also be detrimental to other deep models or even traditional shallow classifiers.

In our future work, we will further investigate the influence of adversarial examples on other remote sensing data (e.g., hyperspectral image and LiDAR data), and try to increase the resistibility of deep models toward adversarial examples for remote sensing tasks.

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